

Econ 139: Labor Economics

Trade (Empirics)

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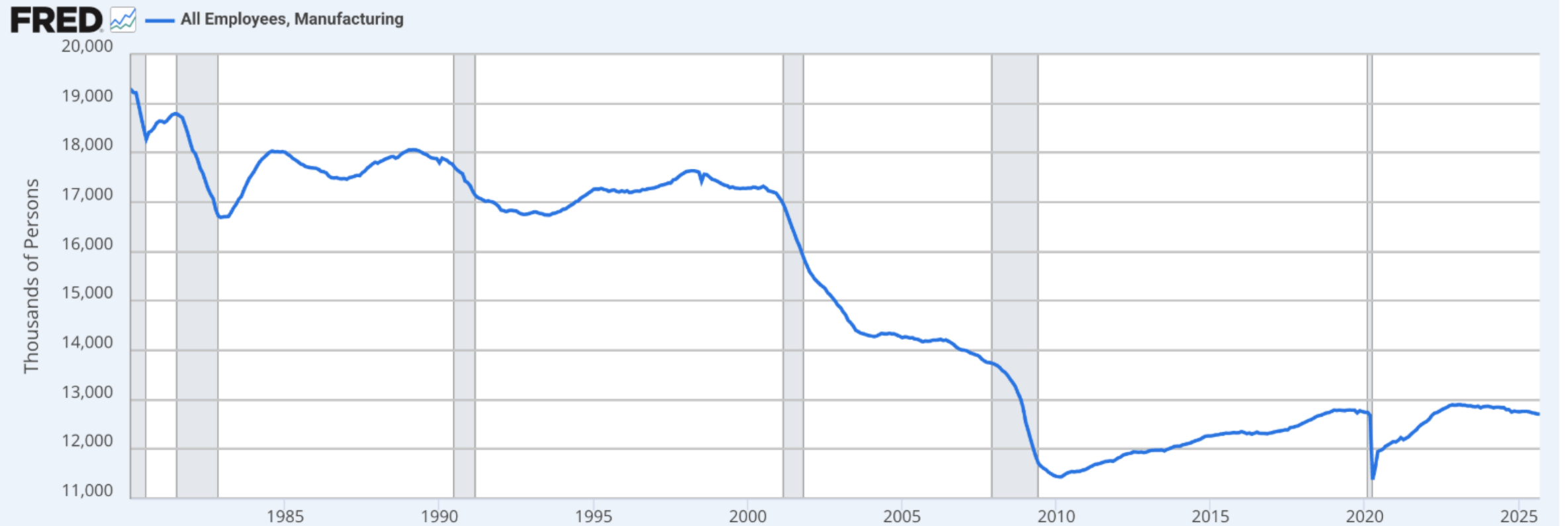
Trade (Empirics)

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- There are two trends that have moved in parallel

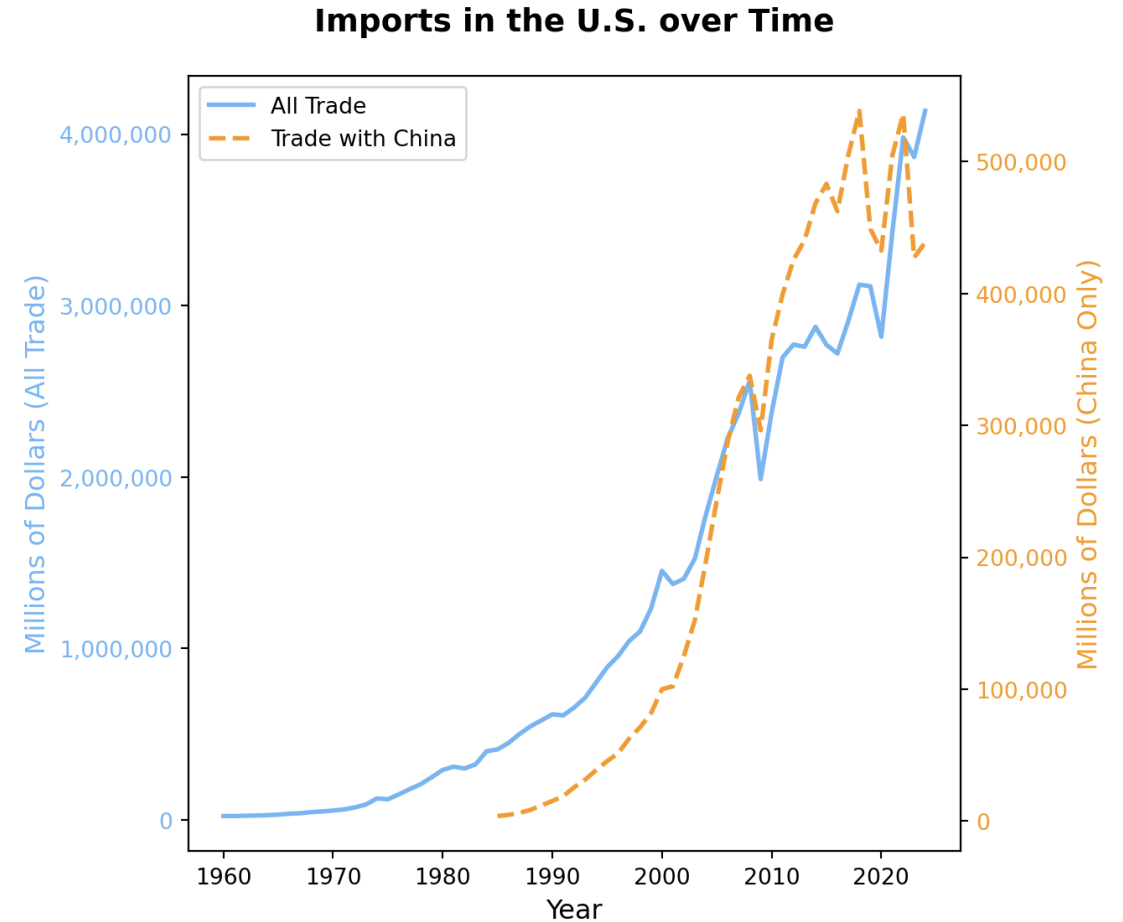
Trade (Empirics)

- Much of the recent focus in trade (from the US perspective) has been on the increase in trade with China
- There are two trends that have moved in parallel
 - Decline in US manufacturing
 - Rising imports from China

Declining Manufacturing Employment



Increase Trade from China



Unanswered Questions

- Two unanswered questions about these trends:
 - Unclear whether the decline in manufacturing is actually driven by trade from China. We already saw the robot adoption rapidly increased during this period as well
 - What is the impact on manufacturing workers? If they can easily transition jobs, then maybe this is not harming workers much

Autor, Dorn and Hanson (2013)

- **Research Question:** What are the local labor market effects of increased trade with China
- **Research Design:** Compare commuting zones that tend to produce manufacturing goods that faced increased competition with China to commuting zones that faced less increased competition with China

Winners and Losers

- In many standard models of labor markets, we might not expect to see regional effects of trade? Why?
 - **Job mobility:** Workers can move across areas. So if wages fall in areas exposed to trade, workers can move out, which will tend to equalize wages across space
- But whether this actually happens in reality is an empirical question. To answer this question we will look to Autor, Dorn and Hanson (2013)

Autor, Dorn and Hanson (2013)

- To understand the approach in Autor, Dorn and Hanson (2013) we are going to dramatically simplify the approach initially (i.e. this is not what the paper does, but is an illustrative example of the important concepts)
- Assume there is just one industry that saw increased competition from China: manufacturing of electronic computers
- Our goal next is to understand how regions are differentially exposed to this increased trade from China in this industry

Part 1: How Big is the Shock

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 - 2000: value of US imports in this industry was 2 billion
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Part 1: How Big is the Shock

- How impactful an increase in trade in an industry is depends clearly on the amount of trade
- For example, in manufacturing of electronic computers
 - 2000: value of US imports in this industry was 2 billion
 - 2007: value of US imports in this industry was 27 billion
- The change in imports is therefore equal to **25 billion dollars**

Part 2: Which regions are impacted by the shock?

- Let's say manufacturing of electronic computers is done in 3 places in 2000:
 - San Jose, CA – 50% of all manufacturing
 - Austin, TX – 30% of all manufacturing
 - Boston, MA – 20% of all manufacturing
- Then of the 25 billion increase, we will attribute 50% to San Jose, 30% to Texas, and 20% to Boston
- In other words, places with more manufacturing of electronic computers will be more impacted by the increase in trade with China during this time

Part 2: Which regions are impacted by the shock?

Commuting Zone	Share of Electronic Computer Manufacturing (1991)	Apportioned Increase in Imports
San Jose, CA	50%	\$12.5 billion
Austin, TX	30%	\$7.5 billion
Boston, MA	20%	\$5.0 billion
Total	100%	\$25.0 billion

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Last step: scale by the number of workers in the area. A 5 billion dollar shock, for example, will be larger if there aren't that many workers in the area

Part 2: Which regions are impacted by the shock?

Commuting Zone	Total Workers	Apportioned Increase in Imports	Import Exposure per Worker
San Jose, CA	1,000,000	\$12.5 billion	\$12,500
Austin, TX	500,000	\$7.5 billion	\$15,000
Boston, MA	500,000	\$5.0 billion	\$10,000

Generalizing

- Autor, Dorn and Hanson (2013) do this procedure, but for every single industry and occupation
 - They then sum up across all the industries to get a commuting zones exposure per worker
 - There are some additional details we aren't going to cover, but this is the basic intuition of their design

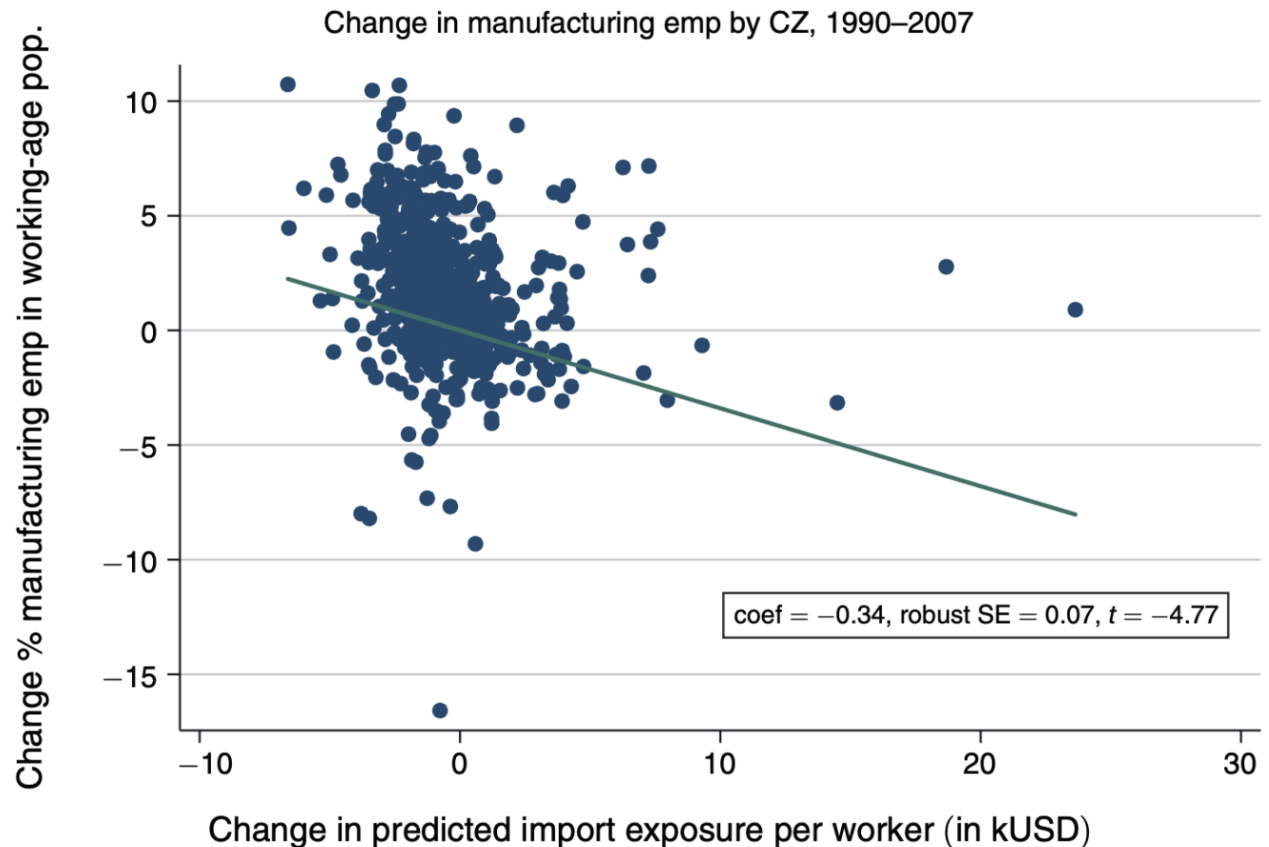
What is the Outcome?

- We now have our measure of exposure to trade
- But what is our outcome? What are we studying the effect of trade on?
 - To start we will look at the **percent of individuals employed in the manufacturing**
 - This will tell us how differential exposure to trade with China is associated with declines in manufacturing

Interpretation

- X-axis is import exposure per 1,000 dollars
 - Value of 10 indicates 10,000 per worker
- Additional 1,000 dollars per worker decreases employment manufacturing by -0.34 percentage points

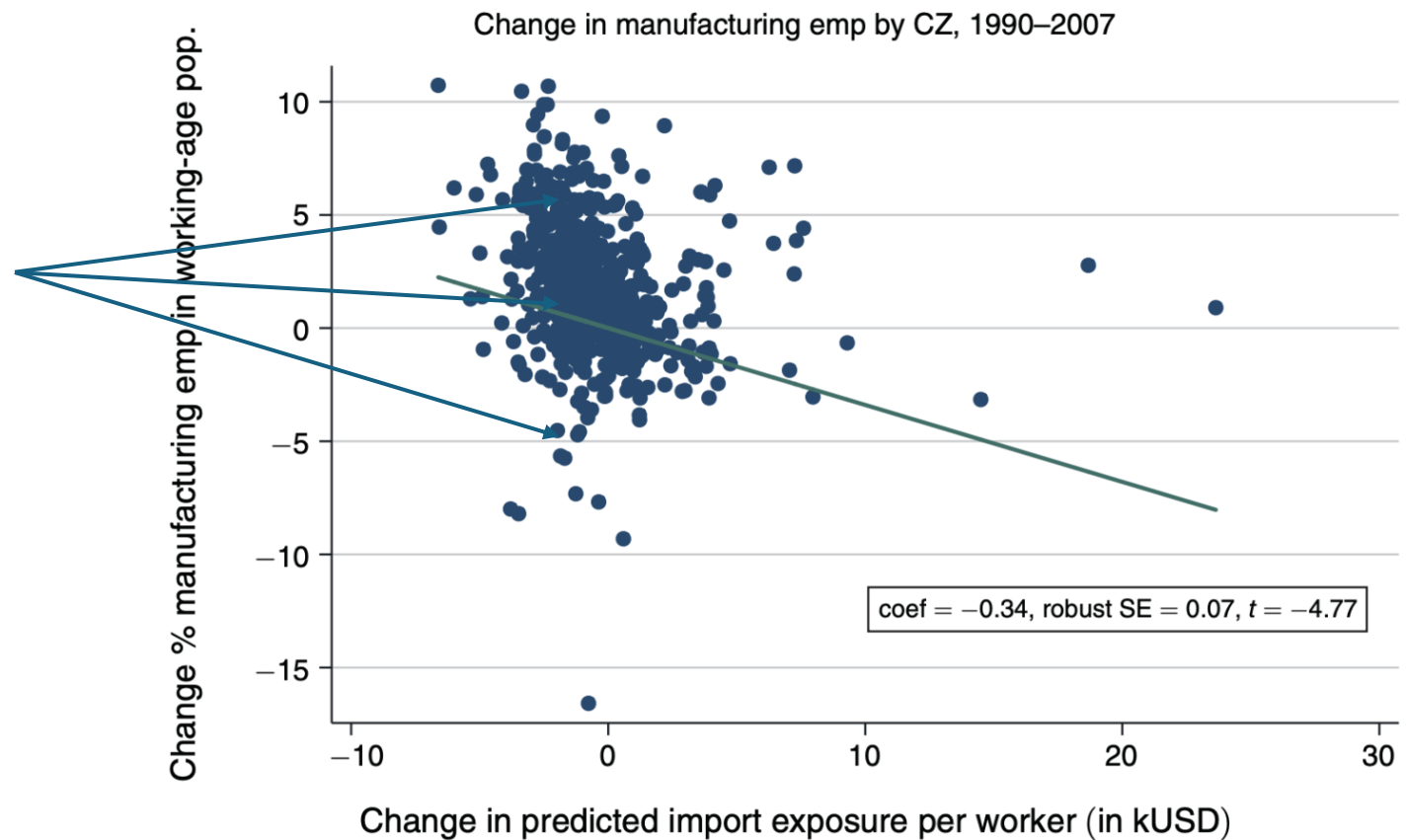
Panel B. OLS reduced form regression, full sample



Interpretation

- Lots of variation not explained by import exposure
- Many places with same level of exposure but very different changes in manufacturing employment

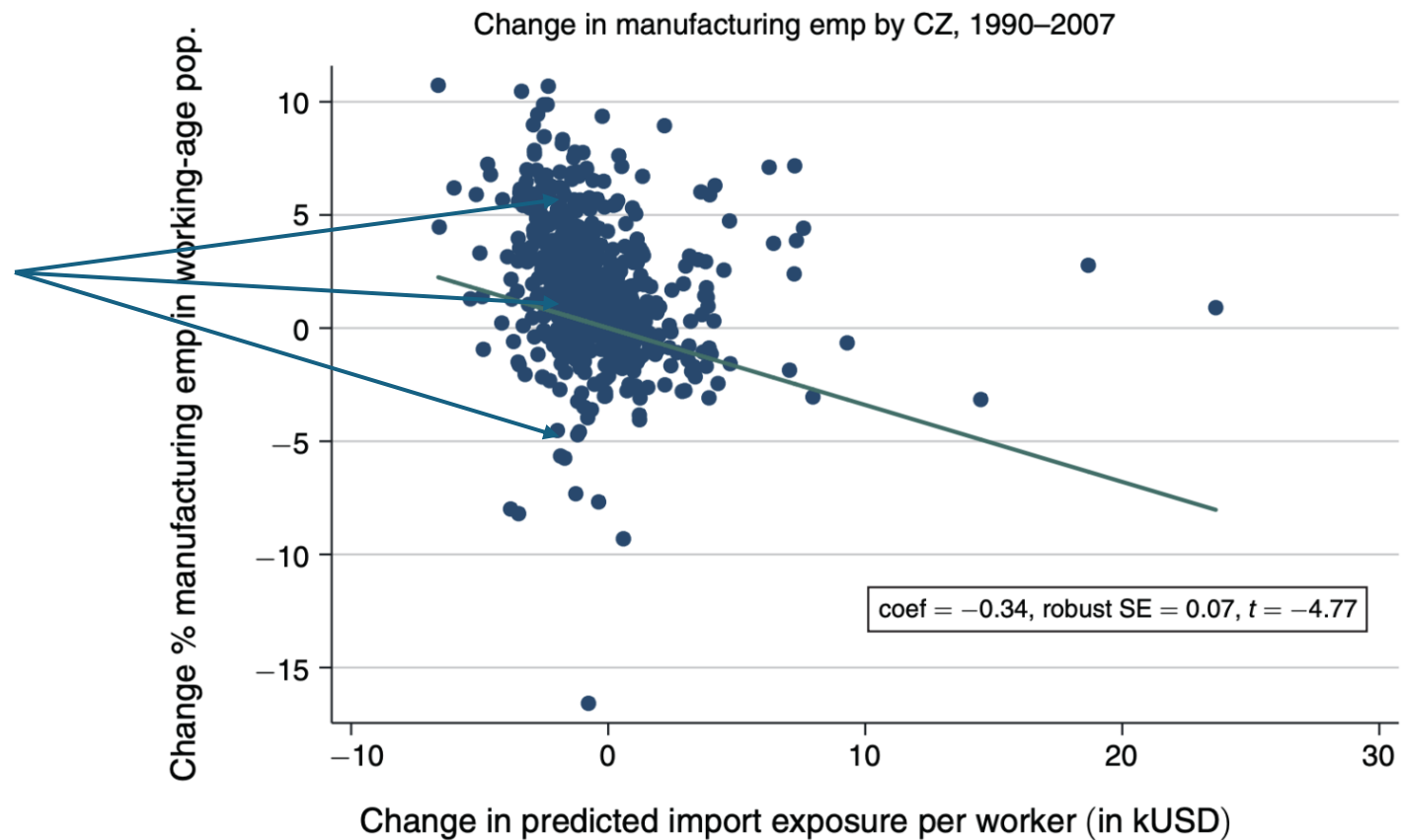
Panel B. OLS reduced form regression, full sample



Interpretation

- Even if there is a drop in manufacturing employment due to trade, is this bad?
 - Do workers move to other locations?
 - Can they find replacement jobs quickly?

Panel B. OLS reduced form regression, full sample



Additional Outcomes

- **Impact on moving:** no evidence that workers in trade-exposed areas move out
 - This is a key mechanism why some thought there may be limited effects of trade. If workers are highly mobile, they can move to areas with more opportunities
- **Impact on unemployment and labor force participation:**
 - A \$1,000 per worker import shock increases the number of unemployed and nonparticipating individuals by 4.9 and 2.1 percent, respectively.

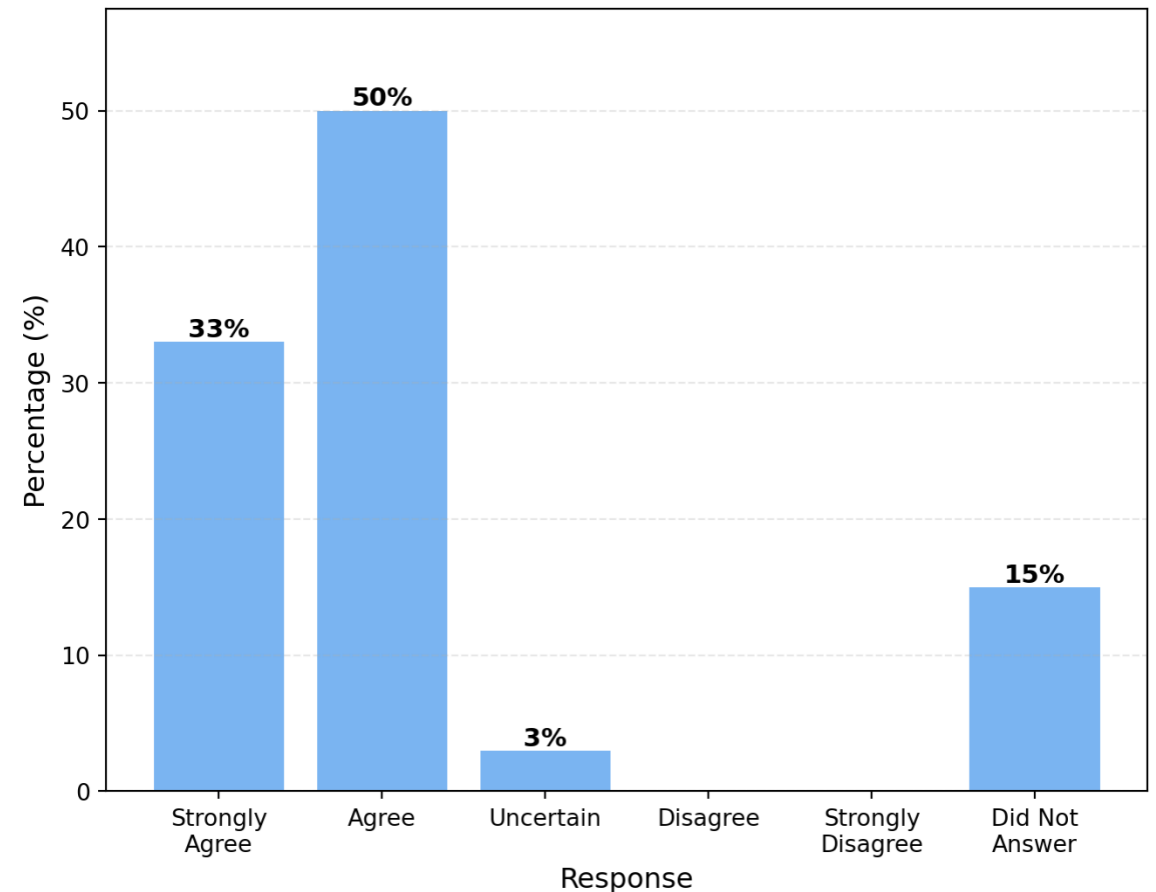
Is this a surprise?

- It was generally assumed that trade can lead to winners and losers
- For example, in 2012, Clark Center asked economists
- *Some Americans who work in the production of competing goods, such as clothing and furniture, are made worse off by trade with China.*

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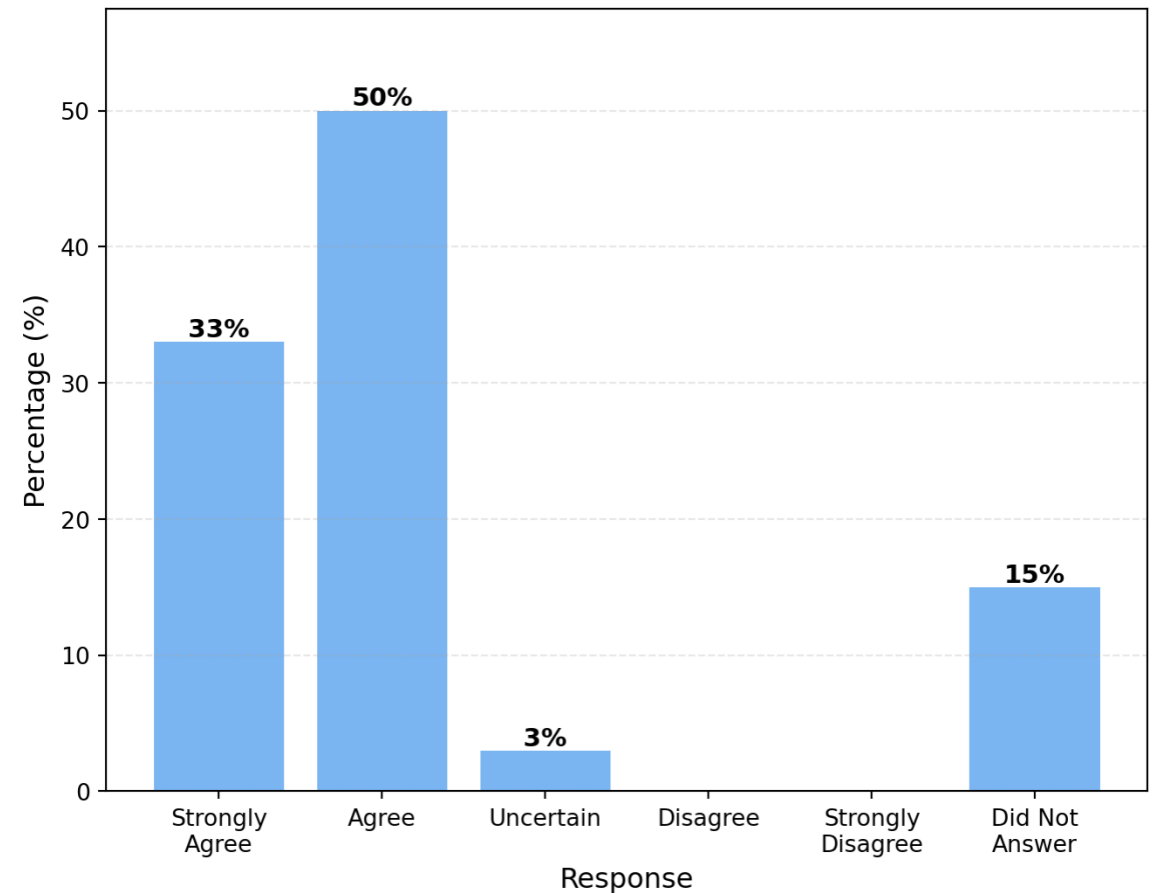
Survey Responses on Trade Impact on American Workers



Is this a surprise?

- What may be surprising is how “local” the effects are
 - Some workers that become unemployed for long periods of time or drop out of the labor force may be much worse off

Survey Responses on Trade Impact on American Workers



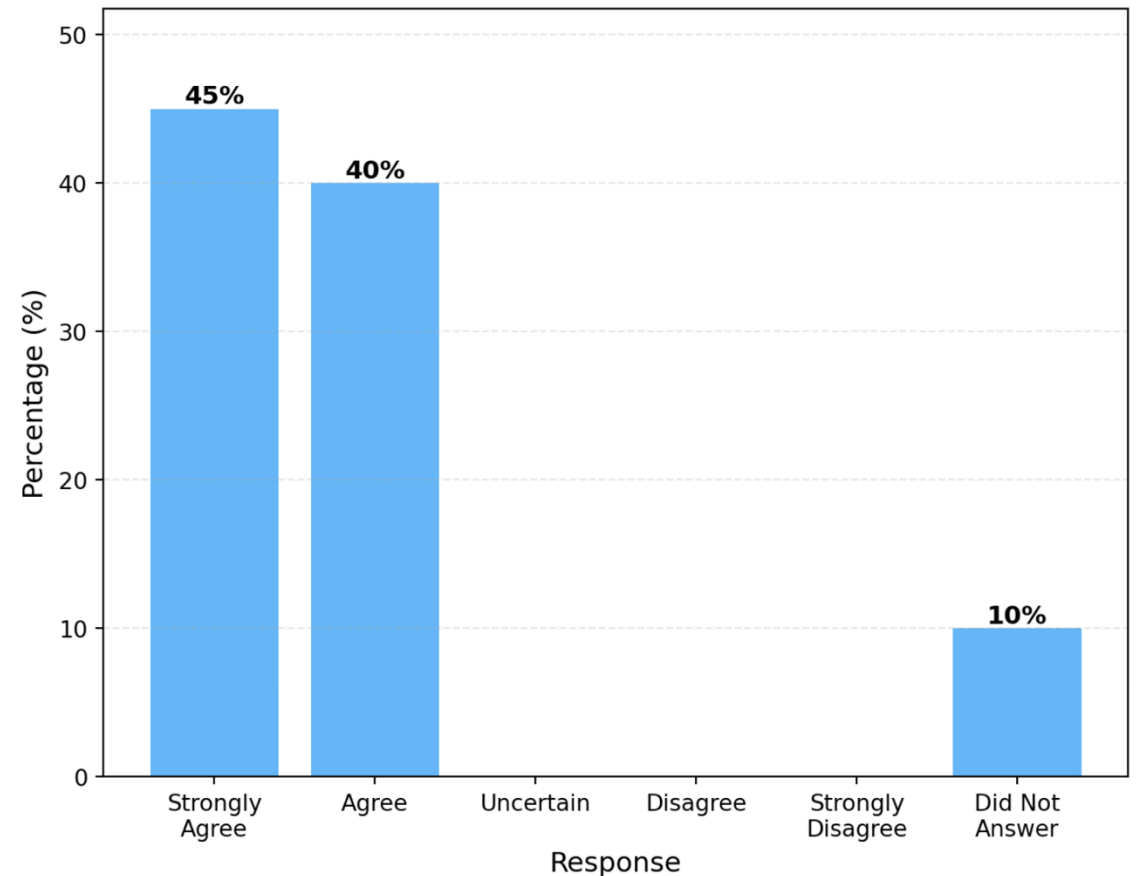
Does this change economists mind on trade?

- The next question in the Clark Center asked was specifically about trade with China
- *Trade with China makes most Americans better off because, among other advantages, they can buy goods that are made or assembled more cheaply in China.*

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Survey Responses on Trade with China Being Beneficial



Incorporating Benefits of Trade

- The Autor, Dorn and Hanson (2013) is an extremely influential paper showing the effects of trade with China on manufacturing workers
- However, it shows the costs (i.e. job losses for workers in manufacturing), not the benefits
 - Benefits could be consumer-welfare from purchasing cheaper goods
 - But increases in disposable income may also lead to more spending elsewhere in the economy, leading to jobs in other sectors
- These benefits are sometimes hard to identify. Harder to figure out a design that would allow us to identify the impact of trade on national income.

Incorporating Benefits of Trade

- Bernhofen and Brown (2005) study the unique situation of Japan, which opened to trade in the early 1850s, coming from complete autarky
 - The situation in Japan was somewhat unique. They closed to trade in 1639 due to religious concern over Portuguese attempts to Christianize the Japanese population
 - They were forced to re-open to trade by military threat from the U.S., eventually signing trade treaties that opened Japan in 1859
 - Bernhofen and Brown (2005) find gains from trade equal of no more than 9 percent

Incorporating Benefits of Trade

- Another popular approach is to use differences in the cost of trade across countries to identify impacts of trade on income
 - Ex: some countries tend to be located outside major shipping lanes, making it more difficult to trade

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- Another popular approach is to use differences in the cost of trade across countries to identify impacts of trade on income
 - Ex: some countries tend to be located outside major shipping lanes, making it more difficult to trade
- Feyrer (2019) uses a clever idea: air freight dramatically shifted costs for some countries relative to others
- Did countries whose trade costs decreased due to air freight see large increases in income
 - Yes, Differences in predicted trade growth can explain roughly 17 percent of the variation in cross-country income growth between 1960 and 1995.

Conclusion

- Key concepts in trade
 - Comparative advantage
 - Distributional effects – trade has winners and losers. In the US, rise in trade with China led to large losses for workers in certain manufacturing industries
 - Aggregate effects – overall, most economists tend to think trade is beneficial overall, but policies may be needed to compensate those who are negatively impacted